

ENDOCRINOLOGIC DISORDERS

Diabetes Mellitus

Objectives

Upon completion of the session, you will be able to:

1. Distinguish type 1, type 2, and gestational diabetes.
2. List screening and diagnostic criteria for diabetes.
3. Set therapeutic goals for blood glucose for DM patients.
4. Recommend pharmacologic therapies for patients with T1 and T2DM
5. Recommend pharmacologic management of acute and chronic complications of DM
6. Develop therapeutic monitoring plan for a patient with diabetes

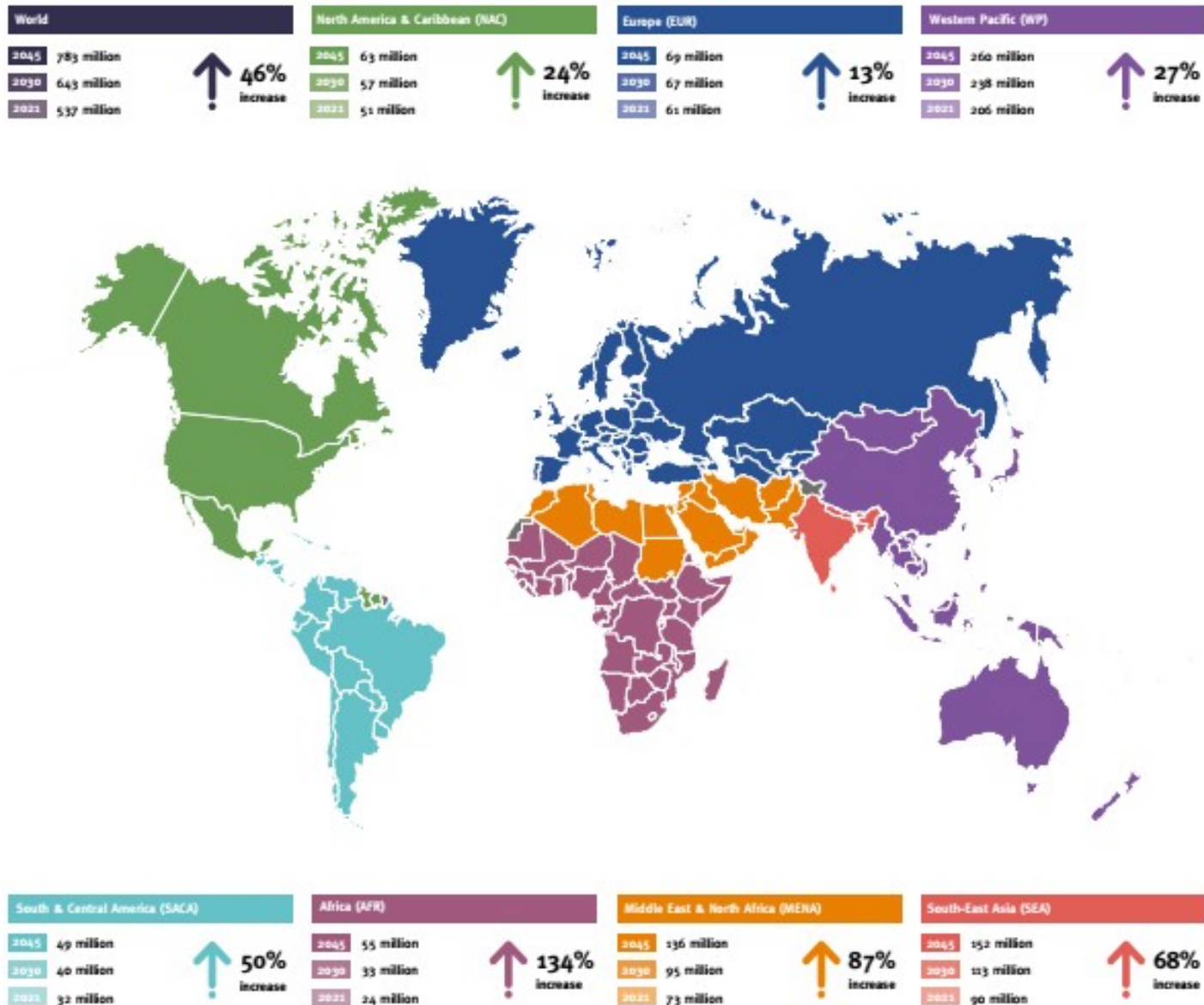
Definition

- Diabetes mellitus (DM) is a heterogeneous group of metabolic disorders characterized by **hyperglycemia**.
- It is associated with abnormalities in **carbohydrate, fat, and protein metabolism** and may result in chronic complications including
- DM is characterized by a ***complete lack of insulin***, a ***relative lack of insulin***, or ***insulin resistance*** as well as disorders of other hormones.
- These defects result in an inability to use glucose for energy

Epidemiology

- The global prevalence of DM among adults over 18 years was **8.5% in 2014**
- Globally 425 million people with DM in 2017
- The total impact of DM in 2012 was approximately \$245 billion in US,
 - ✍ Direct medical costs: \$176 billion
 - ✍ Indirect costs: \$69 billion (secondary to disability, work loss, and premature mortality)
- **Ethiopia: 3.2% of the adult population (2015)**
- **Future Projections**
- The IDF projects a significant increase in diabetes cases in Ethiopia:
- **2030:** Approximately 2,740,800 adults (5.0% prevalence).
- **2045:** An estimated 4,755,500 adults (5.3% prevalence)

EPIDEMIOLOGY

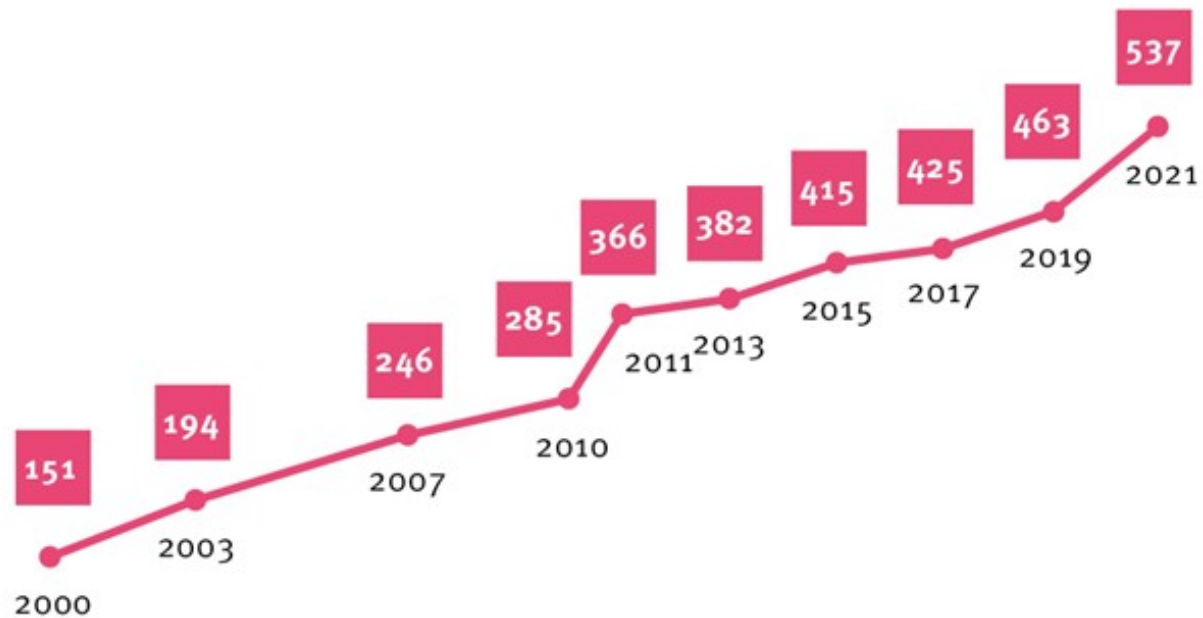


EPIDEMIOLOGY

Top 10 countries for number of adults with diabetes (20-79 years) in 2021 and 2045

2021			2045		
Rank	Country	Number of people with diabetes (millions)	Rank	Country	Number of people with diabetes (millions)
1	China	140.9	1	China	174.4
2	India	74.2	2	India	124.9
3	Pakistan	33.0	3	Pakistan	62.2
4	USA	32.2	4	USA	36.3
5	Indonesia	19.5	5	Indonesia	28.6
6	Brasil	15.7	6	Brazil	23.2
7	Mexico	14.1	7	Bangladesh	22.3
8	Bangladesh	13.1	8	Mexico	21.2
9	Japan	11.0	9	Egypt	20.0
10	Egypt	10.9	10	Japan	13.4

Estimates of global prevalence of diabetes in adults (20-79 years) from 2000 to 2021



Epidemiology

- The increasing prevalence of DM is partly caused by three influences:
 - **Lifestyle:** A sedentary lifestyle coupled with greater consumption of high-fat, high-carbohydrate foods, and larger portion sizes
 - **Ethnicity:** Native Americans, Hispanic Americans, Asian Americans, African Americans, and Pacific Islanders.
 - **Age:** As the population ages, incidence of T2DM is expected to increase

Classification of Diabetes

- The vast majority of diabetic patients are classified into one of two broad categories:
 - ✍ **Type 1 diabetes** caused by an absolute deficiency of insulin or
 - ✍ ~65% of pediatric patients with DM; 5 – 10% of adults with DM.
 - ✍ **Type 2 diabetes** defined by the presence of insulin resistance with an inadequate compensatory increase in insulin secretion.
 - ✍ ~35% of pediatric patients with DM; 90% to 95% of adults with DM.

Etiologic Classification of DM

1. **Type 1 diabetes** (β -cell destruction, usually leading to **absolute insulin deficiency**)

↳ Immune mediated

↳ Idiopathic

2. **Type 2 diabetes**

↳ **Insulin resistance** with relative insulin deficiency: Early stage of T2 DM.

↳ Insulin secretory defect with insulin resistance: Later stage of T2 DM.

3. **Other specific types**

↳ Genetic defects of β -cell function

↳ Diseases of the exocrine pancreas (pancreatitis, trauma, pancreatectomy, neoplasia, cystic fibrosis)

↳ Endocrinopathies (Acromegaly, cushing's syndrome, glucagonoma, pheochromocytoma, hyperthyroidism, somatostatinoma, aldosteronoma)

↳ Drug- or chemical-induced: pentamidine, nicotinic acid, glucocorticoids, thyroid hormone, β -adrenergic agonists, thiazides, phenytoin, interferon alpha

4. **Gestational diabetes mellitus (GDM)**

Pathogenesis

Type I DM

- Type I DM results from **pancreatic β -cell failure**: with “**absolute**” deficiency of insulin secretion.
 -  Immune-mediated destruction of pancreatic β cells,
 -  Unknown or idiopathic processes (rare).
- There often is a **long preclinical period of immune-mediated β -cell destruction** later followed by onset of hyperglycemia when **80%** to **90%** of the β cells have been destroyed.
- Occasionally there is a period of **transient remission** called the “**honeymoon**”

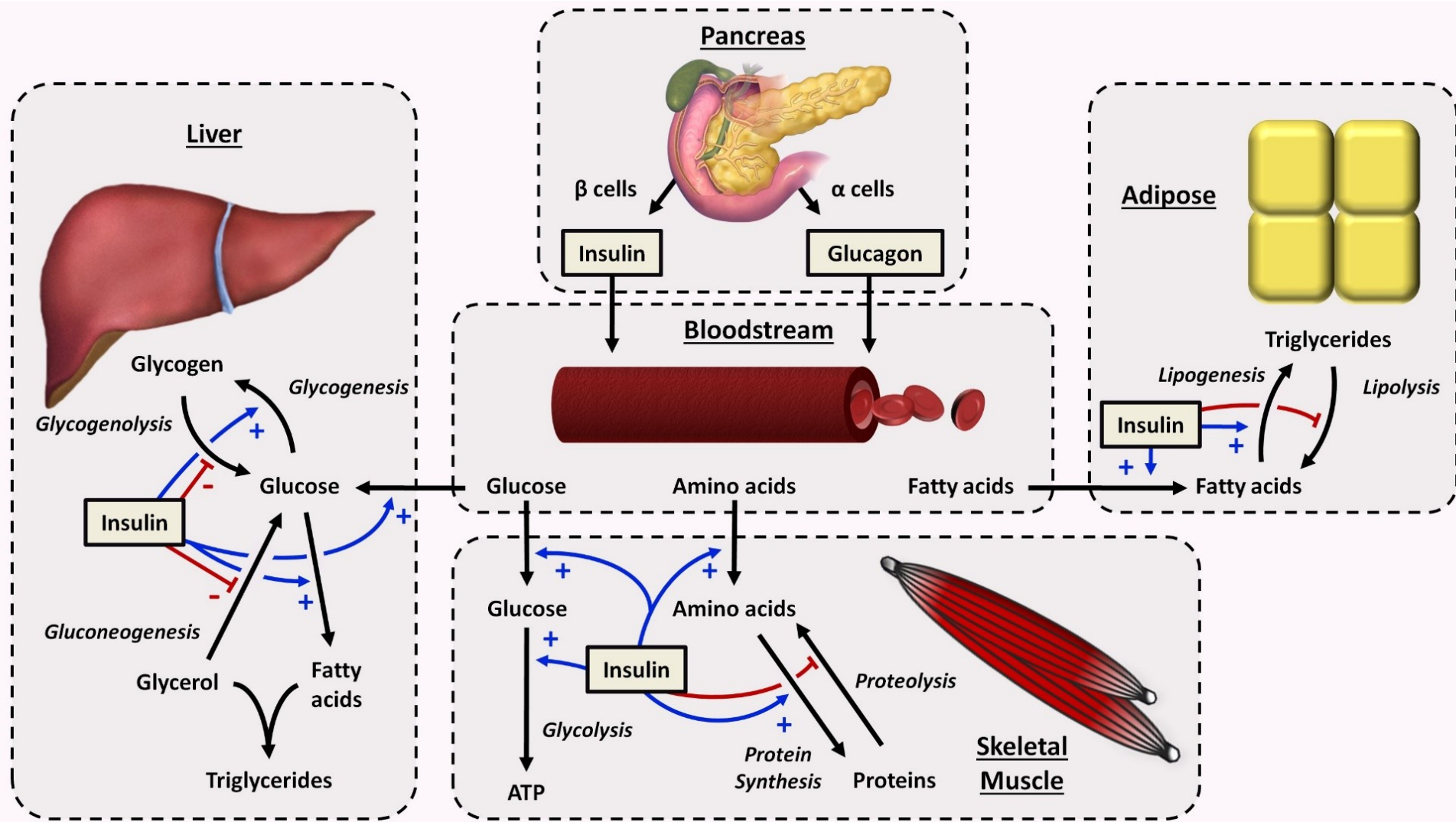
Pathogenesis

- The autoimmune process is mediated by macrophages and T lymphocytes with circulating autoantibodies to various β -cell antigens.
- The most commonly detected antibody associated with type I DM is the islet cell antibodies (ICA)
- β -Cell autoimmunity may precede the diagnosis of type I DM by up to 9 to 13 years.
- Autoimmunity may remit in some individuals, or can progress to absolute β -cell failure in others.
- Other autoimmune disorders frequently associated with type I DM include Hashimoto's thyroiditis, Graves' disease, Addison's disease, vitiligo, and celiac sprue.

Pathogenesis

- The autoimmune destruction of pancreatic β -cell function results in **hyperglycemia** due to an **absolute deficiency of insulin**.
- Insulin lowers blood glucose (BG) by a variety of mechanisms, including stimulation of tissue glucose uptake, suppression of glucose production by the liver, and suppression of free fatty acid (FFA) release from fat cells.
- The suppression of FFAs plays an important role in glucose homeostasis.
- Increased levels of FFAs inhibit the uptake of glucose by muscle and stimulate hepatic gluconeogenesis

Pathogenesis



Pathogenesis

Normal Metabolism

- In the fasting state:

↳ **Glucagon** is produced by **pancreatic α cells** to:

✍ Oppose the action of insulin and

✍ Stimulate hepatic **glucose production** and **glycogenolysis**.

↳ **85% of glucose production** is derived from the **liver**, and the remaining is by the **kidney**.

↳ **75% of total body glucose disposal** takes **place in brain**, neurons, and others.

✍ Non-insulin-dependent tissues

↳ The remaining 25% of glucose metabolism takes place in the **liver and muscle**

✍ Insulin dependent tissues.

Pathogenesis

- In the fed state,
 - ✍ Carbohydrate ingestion increases the plasma glucose concentration and
 - ✍ Stimulates **insulin** release from the pancreatic β cells.
 - The resultant hyperinsulinemia
 - (a) Suppresses hepatic glucose production,
 - (b) Stimulates glucose uptake by peripheral tissues,
 - (c) Suppresses glucagon release (in conjunction with incretin hormones).
 - ✍ The majority (80 to 85%) of glucose is taken up by muscle, with only a small amount (4 to 5%) being metabolized by adipocytes

Pathogenesis

Type 2 Diabetes

- Individuals are characterized by multiple defects including
 - (a) Defects in insulin secretion;
 - (b) Insulin resistance involving muscle, liver, and adipocyte;
 - (c) Excess glucagon secretion;
 - (d) Glucagon-like peptide-1 (GLP-1) deficiency and possibly resistance

Impaired Insulin Secretion

- The pancreas in people with a normal-functioning β cell is able to adjust its secretion of insulin to maintain normal plasma glucose levels.
- In nondiabetic individuals, insulin increases in proportion to the severity of the insulin resistance and plasma glucose remains normal.
- Impaired insulin secretion is a hallmark finding in T2DM.
- People with T2DM lose 5% to 7% of β -cell function per year of diabetes.
- The reasons for this loss are likely multifactorial including
 - glucose toxicity;
 - lipotoxicity;
 - insulin resistance;
 - age;
 - genetics;
 - incretin deficiency.

Incretins

- In the type 2 diabetic patient, decreased postprandial insulin secretion is due to both
 - ✎ Impaired pancreatic β -cell function and
 - ✎ Reduced stimulus for insulin secretion from **gut hormones**.
- Increased insulin secretion in response to an oral glucose stimulus: “the incretin effect”
- **GLP-1** and **glucose-dependent polypeptide** (GIP), are responsible for over 90% of the increased insulin secretion seen in response to an **oral glucose load**.
- Both hormones are rapidly inactivated by the enzyme **dipeptidyl peptidase-4 (DPP-4)**
- In type 2 diabetic patients, this “incretin effect” is blunted

Pathogenesis

Insulin Resistance

- **Liver:**

- ✍ Fasting hyperglycemia

- ↳ Increase in fasting hepatic glucose production

- ✍ Postprandial hyperglycemia

- ↳ From the diet and

- ↳ From continued glucose production from the liver.

- **Peripheral (Muscle)**

- ↳ Impaired intracellular insulin signaling: insulin resistance

Pathogenesis

- **Peripheral (Adipocyte)**

- ↳ Insulin resistance

- Lipolysis: release of FFAs from the adipocyte
 - Gluconeogenesis indirectly through glycerol and FFAs

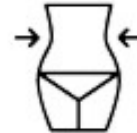
CLINICAL PRESENTATION

Characteristic	Type 1 DM	Type 2 DM
Age	<30 years	>30 years
Onset	Abrupt	Gradual
Body habitus	Lean	Obese or history of obesity
Insulin resistance	Absent	Present
Autoantibodies	Often present	Rarely present
Symptoms	Symptomatic	Often asymptomatic
Ketones at diagnosis	Present	Absent
Need for insulin therapy	Immediate	Years after diagnosis
Acute complications	DKA	HHS
Microvascular complications at diagnosis	No	Common
Macrovascular complications at or before diagnosis	No	Common

Symptoms of type I diabetes



Abnormal thirst and dry mouth



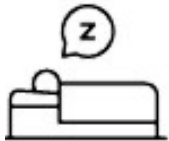
Sudden weight loss



Frequent urination



Bedwetting



Lack of energy, fatigue



Constant hunger



Blurred vision

Symptoms of type 2 diabetes



Excessive thirst and dry mouth



Recurrent fungal infections in the skin



Frequent and abundant urination



Slow healing wounds



Lack of energy, extreme tiredness



Blurred vision



Tingling or numbness in hands and feet

Screening

Type I Diabetes Mellitus

- Screening for type I DM **not** recommended
 - ✍ The prevalence of type I DM is low in the general population.
 - ✍ The acute onset of symptoms in most individuals at time of diagnosis.
- Screening for **β -cell autoantibody status** in high-risk family members may be appropriate

Screening

Type 2 Diabetes Mellitus

- ADA recommends screening
 - ✍ Adults without risk factors starting at age 45 years
 - ↳ Repeat every 3 years if normal
 - ✍ Any age in individuals who are overweight ($\text{BMI} \geq 25 \text{ kg/m}^2$) and have at least one other risk factor
 - ↳ Physical inactivity,
 - ↳ First-degree relative with diabetes or high-risk ethnicity/race,
 - ↳ Women who have delivered a baby $>4 \text{ kg}$ or a history of GDM,
 - ↳ Hypertension,
 - ↳ High triglycerides,
 - ↳ Low HDL,
 - ↳ Women with polycystic ovary syndrome,
 - ↳ A1C is 5.7% or greater, IGT, or IFG in previous testing
 - ↳ Acanthosis nigricans, or
 - ↳ History of cardiovascular disease

Screening

Gestational DM

- ✍ Screen at **first prenatal** visit for undiagnosed type 2 DM
- ✍ Screen at **24–28 weeks' gestation** using a 75-g OGTT.
- ✍ If a diagnosis of **gestational DM** is made, screen for diabetes **6 – 12 weeks** after delivery.
- ✍ Continue to screen patients who have had gestational DM every 3 years for type 2 DM for life

DM Diagnosis

- I. Type 1 and type 2 DM diagnosis
 - a) Glycemic values in nonpregnant patients
 - i. FPG: ≥ 126 mg/dL
 - ✓ Easy and preferred method
 - ii. Random plasma glucose: ≥ 200 mg/dL
 - ✓ With symptoms of hyperglycemia: polyuria, polydipsia, and unexplained weight loss
 - iii. OGTT: ≥ 200 mg/dL
 - ✓ Plasma glucose concentration obtained 2 hours after a 75-g oral glucose ingestion
 - iv. A1C (glycated hemoglobin): $\geq 6.5\%$

DM Diagnosis

b) Gestational diabetes diagnosis: Glycemic values in pregnancy

- 75-g OGTT at 24 – 28 weeks' gestation

 1 hour after OGTT: 180 mg/dL or greater

 2 hours after OGTT: 153 mg/dL or greater

DM Diagnosis

c) Prediabetes diagnosis (high-risk population)

 **IFG**: FPG between 100 and 125 mg/dL

 **IGT**: 2-hour plasma glucose after OGTT (75 g)
between 140 and 199 mg/dL

 **A1C**: between 5.7% and 6.4%

Management of DM

Goals of Diabetes Management

1. Prevent the onset of acute or chronic complications
2. Glycemic therapy goals
 -  A1C less than 6.5%
 -  Obtain every 6 months in patients at goal A1C and every 3 months in those over goal.
 -  FPG: 80–125 mg/dL
 -  Postprandial glucose (1–2 hours after a meal): <200 mg/dL
3. Nonglycemic therapy goals
 -  Blood pressure <140/90 mm Hg

Management of Type I DM

Insulin agents

- Categorized on the basis of duration after injection
 - i. Rapid acting: Insulin aspart, lispro and glulisine
 - ii. Short acting: Regular human insulin
 - iii. Intermediate acting: Neutral protamine Hagedorn (NPH)
 - iv. Long acting: Insulin glargine and detemir; cannot be mixed with other insulins
- Combination insulins (intermediate or long-acting, regular or rapid acting): 70/30, 75/25

Characteristics of U-100 (100 units/mL) Insulins

Category	Drug Name	Clarity	Onset	Inj. Time Before Meal	Peak (hours)	Duration (hours)
Rapid acting	Aspart Lispro Glulisine	Clear	15–30 minutes	15 minutes	1–3	2–5
Short acting	Regular	Clear	30–60 minutes	30	2–3	4–6
Intermediate acting	NPH	Cloudy	1–2 hours	N/A	4–8	10–20
Long acting	Detemir Glargine Degludec	Clear	2–4 hours 1–2 hours 1–2 hours	N/A	6–8 “Peakless” “Peakless”	6–24 ~24 24–42

Management of Type 1 DM

- **Glycemic target**
 - i. Regular and short-acting insulins: target postprandial glucose concentrations
 - ii. Intermediate and long-acting insulins: target fasting glucose concentrations.

Insulin therapy

- Estimate **total daily insulin (TDI)** requirements

-  Weight-based estimate if insulin naïve

- 0.3 – 0.6 unit/kg/day

I. **NPH (2/3) and regular (1/3) insulin**

-  Two-thirds of TDI given before morning meal.

- Two-thirds of this is given as NPH and one third as regular insulin.

-  One-third of TDI given before the evening meal

- Two-thirds of this is given as NPH and one-third as regular insulin

-  **Advantages:** Daily insulin injection frequency two or three times daily and inexpensive

-  **Disadvantages:** Does not **mimic natural insulin secretion** pattern; prone to hypoglycemic events

Insulin therapy

2. Basal/bolus insulin therapy (physiologic insulin therapy)

- ✍ Use of newer insulin analogs to better mimic **natural insulin secretion patterns**
- ✍ Provides day-long **basal insulin** to prevent ketosis and control FPG
 - **NPH insulin** twice daily, **insulin glargine** once daily or **insulin detemir** once or twice daily
 - Basal requirements are 50% of estimated TDI.
- ✍ Provides **bolus insulin** to **control postprandial hyperglycemia**
 - **Rapid-acting insulin** (lispro, aspart, glulisine), **regular insulin**
 - Bolus requirements are 50% of estimated TDI split three ways before meals

Insulin therapy

2. Basal/bolus insulin therapy

- Advantage:
 - ✍ More physiologic,
 - ✍ Less hypoglycemia,
 - ✍ More flexible to patient meal times
- Disadvantage:
 - ✍ Cost
 - ✍ Increased frequency and number of daily injections (rapid-acting and basal insulin must be injected separately)

Insulin therapy

2. Basal/bolus insulin therapy

- **Correctional insulin needs**

- ✍ Always a need to correct for **hyperglycemic excursions**, despite optimal basal/bolus therapy

- ✍ “1800 rule”: $1800/\text{TDI} = \# \text{ mg/dL of glucose lowering per 1 unit of rapid-acting insulin.}$

- For example, if TDI is 60 units, $1800/60 = 30$, suggesting 1 unit of **rapid-acting insulin** will reduce blood glucose concentration by 30 mg/dL.

- ✍ Also referred to as **insulin sensitivity**

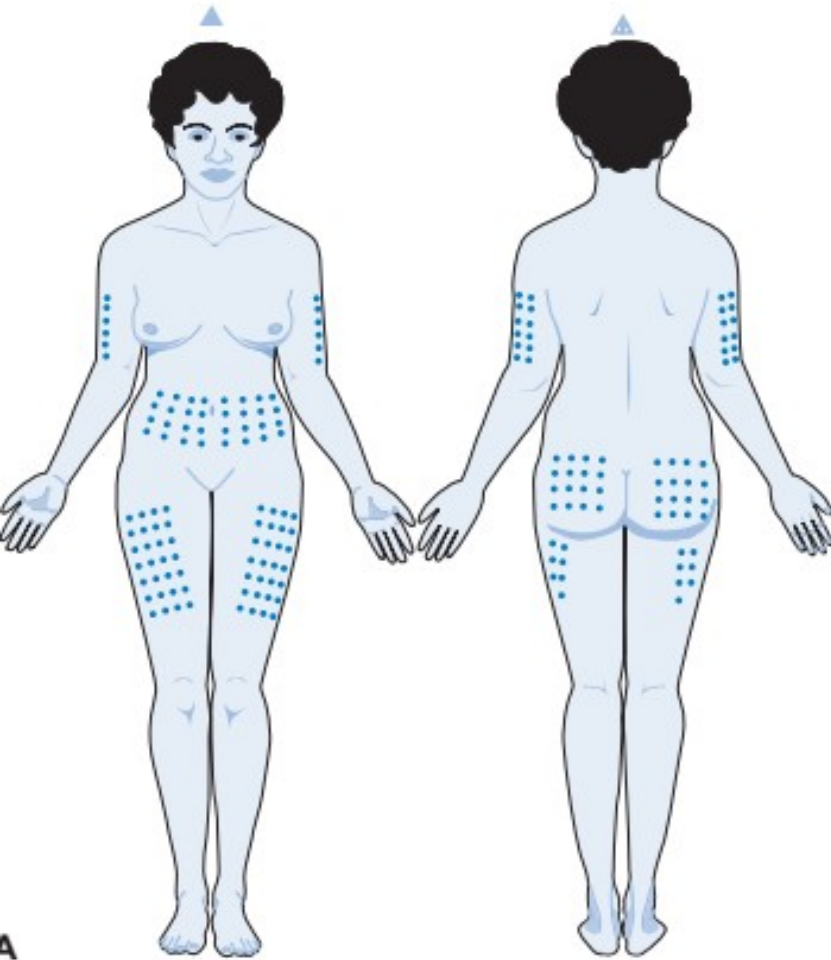
Insulin therapy

3. Continuous subcutaneous insulin infusion (CSII)

Insulin pump

-  Device allows every patient-specific hourly basal dosing and bolus insulin dosing.
-  Uses rapid-acting insulins
-  Requires patient education and carbohydrate counting

Insulin Injection Sites



- Insulin absorption is affected by
 - 1) Site: abdomen $>$ arm, buttock
 - 2) Exercise: increases absorption

Case Study, Type I DM

- A.H., a slender, 18-year-old woman who was recently discharged from the hospital for severe dehydration and mild ketoacidosis, is referred to the Diabetes Clinic. A fasting and a random plasma glucose ordered subsequently were 190 mg/dL and 250 mg/dL. Approximately 4 weeks before she was hospitalized, A.H. had moved from her hometown to Addis Ababa to attend college. This is her first time to move away from home.

Case Study, Type I DM

- In retrospect, she remembers that she had symptoms of **polydipsia**, **nocturia** (six times a night), **fatigue**, and a 5-KG weight loss during this period, which she **attributed to the anxiety associated with her move away from home and adjustment to her new environment**. Her medical history is remarkable for **recurrent upper respiratory infections** and three cases of **vaginal candidiasis** in the past 6 months. Her family history is negative for diabetes, and she takes no medications.

Case Study, Type I DM

- Physical examination is within normal limits. She weighs 50 kg and is 165cm tall. Laboratory results are as follows: FPG, 280 mg/dL; A1C, 14%; and trace **urine ketones**. On the basis of her history and laboratory findings, the presumptive diagnosis is type I diabetes.

Case Study, Type I DM

1. Which findings are consistent with this diagnosis in A.H.?
2. How do you consider insulin therapy for this patient?

Case Study, DM diagnosis

- A 46-year-old man presents for his annual check-up. He states that he has been going to the bathroom more frequently than normal and has lost approximately 9.1 kg in the past 6 months without trying. His random glucose today is 252 mg/dL. Which of the following is most appropriate to confirm diagnosis of type 2 DM?
 - A. Obtain a fasting glucose level in the morning.
 - B. Perform a 2-hour oral glucose tolerance test.
 - C. Obtain patient's hemoglobin A1c.
 - D. Obtain a repeat random glucose level.
 - E. Nothing; there is enough information available to diagnose patient with type 2 DM.

Case Study, Screening DM

A 36-year-old Ethiopian woman is concerned about her risks for developing type 2 diabetes because her mother and father have the disease. Currently, she weighs 66 kg ($\text{BMI} = 26 \text{ kg/m}^2$) and has no other medical conditions. All of her three children weighed less than 3.6 kg at birth. The most appropriate recommendation for screening is to begin:

- A. At the age of 45 then every 3 years if normal.
- B. At the age of 45 then every year if normal.
- C. At the age of 40 then every 3 years if normal.
- D. At the age of 40 then every year if normal.
- E. As soon as possible then every year if normal.

Management of Type 2 DM

-  Metformin
-  Sulfonylureas
-  Meglitinides
-  Thiazolidinediones (TZDs)
-  Dipeptidyl peptidase-4 (DPP-4) inhibitors
-  Sodium glucose cotransporter-2 (SGLT-2) inhibitors
-  GLP-1 agonists
-  Insulin

Metformin (Biguanide)

- **Mechanism**

- ✍ Reduces hepatic gluconeogenesis
- ✍ enhances insulin sensitivity in peripheral tissues, allowing for increased glucose uptake into muscle cells and to a lesser extent, intestinal absorption of glucose

- **Dosing**

- ✍ Initial: 500 mg once or twice daily (once daily with extended-release formulation)
- ✍ Maximal daily dose: 2550 mg (more commonly, 2000 mg/day)
- ✍ Can increase at weekly intervals as necessary

Metformin

- **Adverse effects**

- ✍ Common: Nausea, vomiting, diarrhea, epigastric pain
- ✍ Less common: Decrease in vitamin B12 level, lactic acidosis (rare)
- ✍ Signs or symptoms of lactic acidosis: acidosis, nausea, vomiting, diarrhea, increased RR, abdominal pain, shock, and tachycardia

- **Contraindications and precautions (because of risk of lactic acidosis)**

- ✍ Renal impairment(Contraindicated: **eGFR < 30 mL/min**)
- ✍ Age 80 years or older
- ✍ High risk of cardiovascular event or hypoxic state
- ✍ Hepatic impairment
- ✍ Congestive heart failure

Metformin

- **Efficacy**
- Reduce A1C by 1%–2% & FPG by 60–80 mg/dL in drug-naïve patients with initial A1C of approximately 9%
 - ✍ Some benefit in triglyceride, LDL reduction, HDL increase(2%) and minimal weight loss
 - ✍ Considered first-line therapy

Sulfonylureas

- Bind to receptor (sulfonylurea receptor (SUR I) on **pancreatic β cells**, leading to **membrane depolarization**, with subsequent stimulation of insulin secretion (insulin secretagogue)
- **1st generation agents** seldom used today (chlorpropamide, tolbutamide, tolazamide)
- **2nd generation agents** (glyburide(Glibenclamide), glipizide, glimepiride)
- Dosage titration: Can increase at weekly intervals as necessary

Sulfonylureas

Adverse effects

- Common:
- **Hypoglycemia** (high risk: skipping meals, exercising vigorously, or losing substantial amount of weight, older patients, renal insufficiency),
- **Weight gain**
 - ✍ Less common: Rash, headache, nausea, vomiting, photosensitivity
- **Contraindications and precautions**
 - ✍ Hypersensitivity to sulfonamides
 - ✍ Patients with hypoglycemic unawareness
 - ✍ **Poor renal function** (glipizide may be a better option than glyburide or glimepiride)
- **Efficacy**
- 1 – 2% A1C reduction with FPG reductions of 60–70 mg/dL in drug-naïve patients

Second-Generation Sulfonylurea Dosing

Drug	Initial Dosage	Maximal Daily Dosage (mg)
Glyburide (nonmicronized)	2.5 – 5.0 mg once or twice daily	20
Glyburide (micronized)	1.5 – 3 mg once or twice daily	12
Glipizide	5 mg once or twice daily (once daily with extended release)	40
Glimepiride	1–2 mg once daily	8

Meglitinides

- Stimulate insulin secretion from pancreatic β -cells by binding to a site adjacent to the SUR1 receptor
- As monotherapy, reduce PPG excursions and reduce A1C(0.8%–1%)
- Dosing

Repaglinide

- Initial: 0.5 – 1 mg, take 15 minutes before meals
- Maximal daily dosage: 16 mg

Nateglinide

- 120 mg before meals
- 60 mg if A1C near goal
- Adverse effects: Hypoglycemia and weight gain

Thiazolidinediones (TZDs) (Glitazones)

- Increases expression of genes responsible for glucose metabolism, resulting in **improved insulin sensitivity** in muscle, liver, and fat tissues
- Maximum effects may not be seen until 3–4 months of therapy
- Dosing

Pioglitazone

- Initial: 15 mg once daily
- Maximal daily dosage: 45 mg
- Has cardiovascular benefits (↓ risk of stroke & MI)

Rosiglitazone

- Initial: 4 mg daily as a single daily dose or in divided doses twice daily, increase to 8 mg

TZDs (Glitazones)

- Adverse effects

- ✍ Weight gain (results from both fluid retention & fat accumulation)
- ✍ Fluid retention (due to peripheral vasodilation & improved insulin sensitization in the kidney with increased sodium & water retention)
- ✍ Risk of proximal bone fractures
- ✍ Possible risk of bladder cancer (pioglitazone)
- ✍ Increased risk of heart failure

- Contraindications and precautions

- ✍ Hepatic impairment
- ✍ Class III/IV heart failure (symptomatic heart failure)
- ✍ Existing fluid retention

DPP-4 inhibitors

- Inhibit the breakdown of **GLP-I** secreted during meals,
 - ✍ Increase pancreatic insulin secretion,
 - ✍ Limits glucagon secretion,
 - ✍ Slows gastric emptying, and
 - ✍ Promotes satiety

DPP-4 inhibitors

- **Dosing**

- ✍ **Sitagliptin:** 100 mg once daily

- ↳ CrCl 30 – 50 mL/min: 50 mg once daily

- ↳ CrCl < 30 mL/min: 25 mg once daily

- ✍ **Saxagliptin:** 5 mg once daily

- ↳ CrCl ≤ 50 mL/min: 2.5 mg once daily.

- ↳ When coadministered with strong CYP3A4/5 inhibitor (e.g., ketoconazole): 2.5 mg once daily

- ✍ **Linagliptin:** 5 mg once daily (no dosage adjustment for renal impairment)

- ✍ **Alogliptin:** 25 mg once daily

- ↳ CrCl 30 – 60 mL/min: 12.5 mg once daily

- ↳ CrCl <30 mL/min: 6.25 mg once daily

DPP-4 inhibitors

 **Vildagliptin : 50 mg twice daily**

➤ CrCl < 15 mL/min: 50 mg once daily.

- **Adverse effects**

 Upper respiratory and urinary tract infections,
headache, severe joint pain

- Efficacy: 0.5%–0.8% reduction in A1C,
 - Considered weight neutral and low risk of hypoglycemia

SGLT-2 Inhibitors (Gliflozins)

- Increases urinary glucose excretion by blocking normal reabsorption in the proximal convoluted tubule
- Has some effect on delaying GI glucose absorption
- Reduce A1C by 0.5%–1% and more effective in higher baseline A1C levels
- Can be added to metformin or other second-line agents and can be as monotherapy in patients who cannot tolerate or take metformin
- Recommended for patients at high risk for or with established ASCVD, heart failure, or CKD and those with a need to avoid hypoglycemia or weight gain or loss.
- Unlikely to cause hypoglycemia unless combined with medications such as sulfonylureas, meglitinides, or insulin

SGLT-2 Inhibitors (Gliflozins)

○ Canagliflozin

- 100 mg once daily before the first meal of the day
- Maximal daily dosage: 300 mg
- CrCl 45 - 59 mL/min: 100 mg daily
- Discontinue or do not initiate if eGFR $<45\text{mL/min/L}$

Dapagliflozin

- 5 mg once daily in the morning (with or without food)
- Maximal daily dosage: 10 mg
- Discontinue or do not initiate if eGFR($<60\text{mL/min}$)

Empagliflozin

- 10 mg once daily in the morning (with or without food)
- Maximal daily dosage: 25 mg
- Discontinue or do not initiate if eGFR($<60\text{mL/min/l}$)

SGLT-2 Inhibitors (Gliflozins)

- **Adverse effects**

- ✍ Increased urination
- ✍ Urinary tract infections
- ✍ Genital mycotic infections
- ✍ Polyuria, dehydration, dizziness, or hypotension (due to occur because of the osmotic diuresis effects)
- ✍ Increased hypoglycemia risk with concomitant insulin or insulin secretagogue

- **Contraindications and precautions**

- ✍ Significant renal impairment
- ✍ Suggested to ensure euvolemia before initiating agent

GLP-1 analogs

- Synthetic analog of human GLP-1 that binds to GLP-1 receptors, resulting in:
 - ↳ Glucose-dependent insulin secretion,
 - ↳ Reduction in glucagon secretion
 - ↳ Decreased hepatic glucose output.
 - ↳ Reduced gastric emptying
 - ↳ Promotes satiety
 - ↳ Cause weight loss

GLP-1 analogs

- **Exenatide:**

-  Twice-daily formulation (5 mcg SC twice daily)
-  Once-weekly formulation (2 mg SC once weekly)
-  Avoid if eGFR <30ml/min

- **Liraglutide:**

-  0.6 to 1.2 mg SC once daily

- **Albiglutide:**

-  30 mg SC once weekly

- **Dulaglutide**

-  0.75 mg SC once weekly

- Use with caution in patients with ESKD

- Semaglutide; 0.25–1 mg, once weekly, No dose adjustment

GLP-1 analogs

- Recommended to reduce the risk of major adverse CV events in adults with T2 DM and established ASCVD and CKD and compelling need to avoid hypoglycaemia or to **avoid weight gain or induce weight loss**
- Used in combination with metformin, sulfonylureas, SGLT-2 inhibitors, and basal insulin
- Not be used in combination with DPP-4 inhibitors due to similar MOA
- **Adverse effects:**
 - Nausea, vomiting, and diarrhea
 - Injection site reactions and hypersensitivity reactions (including anaphylaxis and angioedema), acute pancreatitis

α -Glucosidase inhibitors

- Slow the absorption of glucose from the intestine to the bloodstream by slowing the breakdown of large carbohydrates (sucrose) into smaller absorbable sugars
- Two agents available: **Acarbose** and **miglitol**
- Dosing (both agents dosed similarly)
 - ✍ Initial: 25 mg three times daily at each meal
 - ✍ Maximal daily dosage: 300 mg
- Adverse effects
 - ✍ Flatulence, diarrhea, abdominal pain
 - ✍ Increased liver enzymes observed with high dosages of acarbose

Insulin

- **Initial** insulin therapy:
 - ✓ Use insulin early with any of the following baseline characteristics:
 - ✍ A1C greater than 10%
 - ✍ Glucose greater than 300 – 350 mg/dL
 - ✍ Hyperglycemic symptoms
 - ✍ Presence of **urine ketones**

Insulin

- **Adding insulin** to existing oral DM agents
 - ✍ Common to add a basal insulin regimen to existing oral agents when **hyperglycemia** still not controlled
 - Weight-based dosing: For example, 0.1–0.2 unit/kg/day (higher dosages if significant hyperglycemia exists)
 - ✍ Can add **bolus insulin** to one or more meals if postprandial glucose is of concern
 - ✍ **Insulin secretagogues** should be lowered in dosage or discontinued altogether when bolus insulin added to reduce risk of hypoglycemia
 - ✍ **TZDs** should be lowered or discontinued when basal or bolus insulin is added to regimen because of **increased risk of edema**

Insulin

- **Changing** from oral DM medications to insulin-only management
 - ✍ Because of adverse effects, contraindications, lack of efficacy of oral medications
 - ✍ Can follow **NPH/regular insulin** or **basal/bolus approach** similar to that in T1DM
 - ✍ The TDI requirements in T2DM are usually much higher than in T1DM because of insulin resistance

Antihyperglycemic Therapy Recommendations in Type 2 DM

Healthy eating, weight control, increased physical activity, and diabetes education

Initial drug monotherapy

Efficacy ($\downarrow$ HbA_{1c})...
Hypoglycemia.....
Weight.....
Side effects.....
Costs.....

Metformin
High
Low risk
Neutral / loss
GI/lactic acidosis
Low

Dual Therapy

Efficacy ($\downarrow$ HbA_{1c})...
Hypoglycemia.....
Weight.....
Side effects.....
Costs.....

If individualized HbA_{1c} target not reached, proceed to two-drug combination

Metformin +

SU
High
Moderate risk
Gain
Hypoglycemia
Low

Metformin +

TZD
High
Low risk
Gain
Edema, HF, Bone
Moderate

Metformin +

DPP4i
Intermediate
Low risk
Neutral
GI
High

Metformin +

SGLT2 inhibitor
Intermediate
Low risk
Loss
GU, dehydration
High

Metformin +

GLP1-RA
High
Low risk
Loss
GI
High

Metformin +

Insulin
Highest
High risk
Gain
Hypoglycemia
Variable

If individualized HbA_{1c} target not reached after ~3 months, proceed to three-drug combination
(order not to denote any preference choice dependent on variety of patient- and disease-specific factors)

Triple Therapy

SU+
TZD
or SGLT2i
or DPP4i
or GLP1-RA
or Insulin

TZD+
SU
or SGLT2i
or DPP4i
or GLP1-RA
or Insulin

DPP4i+
SU
or TZD
or SGLT2i
or Insulin

SGLT2i+
SU
or TZD
Or DPP4i
or Insulin

GLP1-RA+
SU
or TZD
or Insulin

Insulin+
TZD
or SGLT2 i
or DPP4i
or GLP1-RA

Combination Injectable Therapy

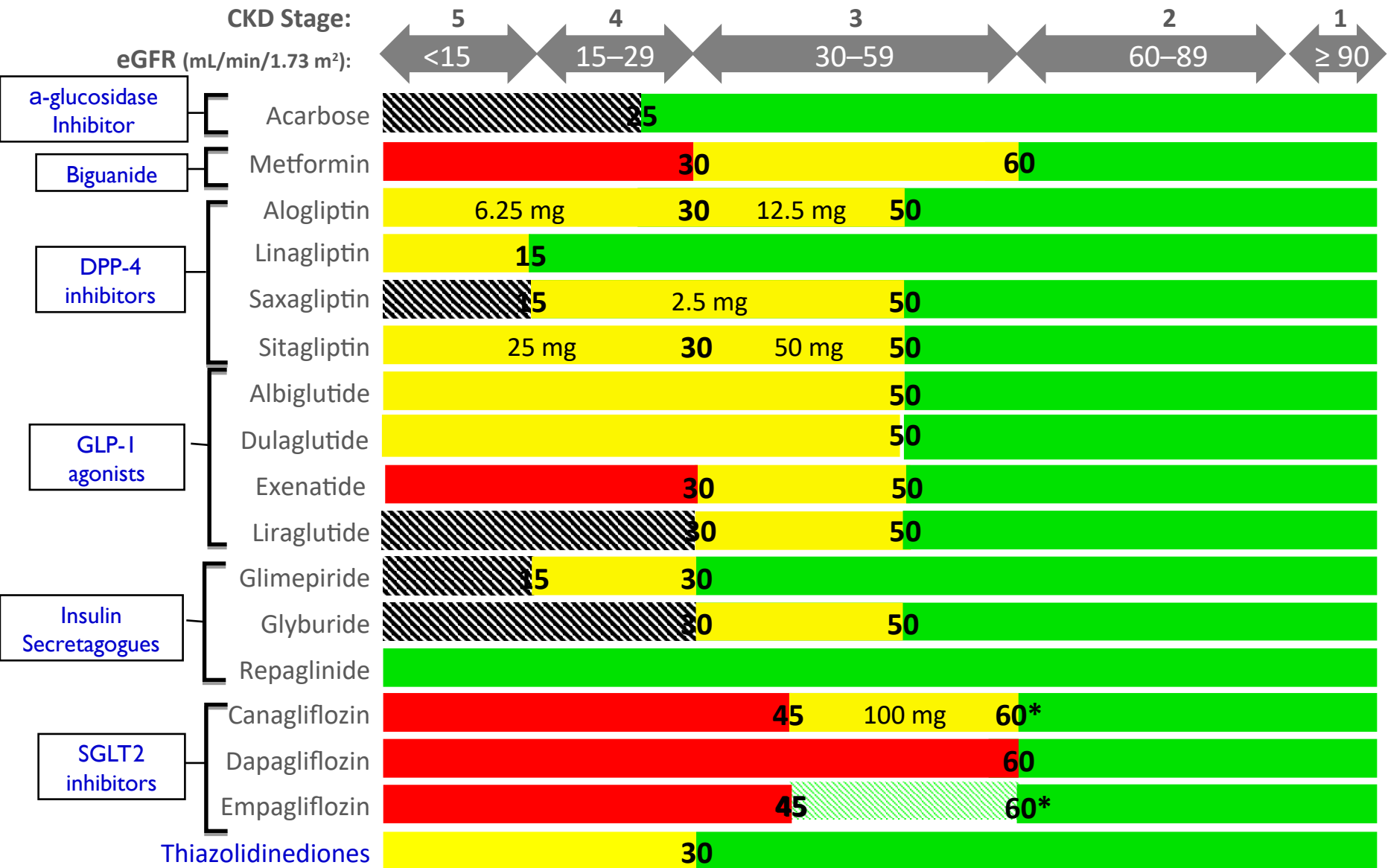
If HbA_{1c} target not achieved after ~3 months of triple therapy and patient (1) on oral therapy, move to injectables; (2) on GLP-1RA, add basal insulin; (3) on optimally titrated basal insulin, add GLP-1RA or mealtime insulin. In refractory patient consider adding TZD or SGLT2i

Basal insulin + Mealtime Insulin or GLP-1 RA

COMPARISON OF ANTIDIABETIC AGENTS

Class	Relative A1C Lowering	Hypoglycemia	Weight	Effect in Cardiovascular Outcome Trial
α -glucosidase inhibitor (acarbose)	↓	Rare	Neutral to ↓	
DPP-4 Inhibitors	↓↓	Rare	Neutral to ↓	alo, saxa, sita: Neutral
GLP-1R agonists	↓↓ to ↓↓↓	Rare	↓↓	lira : Superiority in T2DM patients with clinical CVD lixi : Neutral
Insulin	↓↓↓	Yes	↑↑	Neutral (glar)
Insulin secretagogue: Meglitinide Sulfonylurea	↓↓ ↓↓	Yes Yes	↑ ↑	
SGLT2 inhibitors	↓↓ to ↓↓↓	Rare	↓↓	empa : Superiority in T2DM patients with clinical CVD
Thiazolidinediones	↓↓	Rare	↑↑	Neutral

Antihyperglycemic agents and Renal Function



* = do not initiate if eGFR <60 mL/min

Contraindicated Not recommended Caution and/or reduce dose Safe

No dose adjustment but close monitoring of renal function

Case Study, Type 2 DM

- L.H. is a 45-year-old, overweight woman with central obesity (height, 163cm; weight, 75 KG; BMI, 28.2 kg/m²).
- She was referred to the diabetes clinic when her gynecologist, who had been treating her for recurrent candidial infections, noted glucosuria on routine urinalysis.
- Subsequently, on two separate occasions she was found to have an FPG of 150 mg/dL and 167 mg/dL; an A1C was also checked and it was 8.2%.
- L.H. denies any symptoms of polyphagia or polyuria, although lately she has been more thirsty than usual.
- She does complain of lethargy and takes afternoon naps when she can.

Case Study, Type 2 DM

- L.H.'s other medical problems include **hypertension**, which is well controlled on lisinopril 20 mg/day, and recurrent candidial infections, which are treated with fluconazole.
- She has given birth to four children (birth weights, 3.2, 3.9, 4.5, and 5 Kg) and was told during her last pregnancy that she had “borderline diabetes.”
- She currently works as a loan officer in a local bank and spends her weekends “catching up on her sleep” and reading.
- L.H. has been smoking one pack of cigarettes per day for 20 years and drinks an occasional glass of wine. She drinks at least two regular sodas daily and has a “large” glass of orange juice every morning.

Case Study, Type 2 DM

- Her physical activity consists of routine walking during the day (e.g., to her car).
- Her family history is significant for a sister, aunt, and grandmother with type 2 diabetes; all have “weight problems.” L.H.’s mother is alive and well at age 77; her father died of a heart attack at age 47.
- Fasting laboratory assessment reveals glucose of 147 mg/dL, triglycerides of 400 mg/dL, and an A1C of 8.3%. All other values (including the complete blood count, electrolytes, LFTs, and renal function tests) are within normal limits.
- L.H. is given the diagnosis of type 2 diabetes.

Case Study, Type 2 DM

1. What features in L.H.'s history and physical examination are consistent with this diagnosis?
2. What should the goals of therapy be for L.H. and other patients with type 2 diabetes? Which biochemical indices should be monitored?
3. How should L.H. be managed initially?

GDM & Pregnancy with Preexisting DM

- The adverse outcomes associated with GDM include
 -  Birth defects,
 -  Miscarriage,
 -  Cesarean section delivery
 -  Maternal preeclampsia/eclampsia,
 -  Preterm delivery,
 -  Neonatal hypoglycemia,
 -  Shoulder dystocia,
 -  Birth injury, and
 -  Hyperbilirubinemia

GDM

- **Goals during therapy are**
 - ✍ Preprandial ≤ 95 mg/dL
 - ✍ 1-hour postprandial ≤ 140 mg/dL
 - ✍ 2-hour postprandial ≤ 120 mg/dL
- Insulin is the preferred agent for management of pregestational T1DM and T2DM that are not adequately controlled with diet, exercise, and metformin
- Insulin therapy in the form of **multiple injections** should be used
- For women who are non-adherent to or who refuse insulin, **glyburide or metformin** may be used as alternative agents for glycemic control

MANAGEMENT OF DM COMPLICATIONS

- Acute complications
 -  Hypoglycemia
 -  Diabetic ketoacidosis
 -  HHS
- Chronic complications
 -  Microvascular complications
 -  Macrovascular complications

Hypoglycemia

- Symptoms: anxiousness, sweating, nausea, tachycardia, hunger, clammy skin,
- Blood glucose concentration <60 mg/dL: Patient may or may not be symptomatic.
- Blood glucose <40 mg/dL: Patient is generally symptomatic.
- Blood glucose <20 mg/dL: Can be associated with seizures and coma.

Hypoglycemia

SYMPTOMS

Trembling	Difficulty concentrating
Palpitations	Confusion
Sweating	Weakness
Anxiety	Drowsiness
Hunger	Vision changes
Nausea	Difficulty speaking
	Dizziness

Hypoglycemia

- **Causes**

-  Irregular eating patterns
-  ↑ Physical exercise
-  Gastroparesis (delayed gastric emptying)
-  Defective counter-regulatory responses
-  Excessive dose of insulin or insulin secretagogues (sulfonylureas, glinides)
-  Alcohol ingestion
-  Drugs

Hypoglycemia

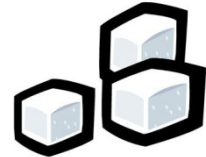
Management

- **Mild to moderate hypoglycemia**
 - ✍ Oral ingestion of 15 g glucose
- **Severe hypoglycemia in a conscious person**
 - ✍ Oral ingestion of 20 g of glucose
- **Severe hypoglycemia in an unconscious individual:**
 - ✍ No IV access:
 - ↳ 1 mg of glucagon SC/IM
 - ✍ IV access
 - ↳ **10-25 g glucose** (20-50 cc of D50W) over 1-3 minutes

Hypoglycemia

Simple Carbohydrate to treat hypoglycemia

- 15 g of glucose in the form of **glucose tablets**
- 15 mL (3 teaspoons) or 3 **packets** of sugar dissolved in water
- 175 mL (3/4 cup) of **juice** or regular **soft drink**
- 15 mL (1 tablespoon) of **honey**



Diabetic Ketoacidosis

- Diabetic ketoacidosis is a reversible but potentially life-threatening medical emergency that results from a **relative or absolute deficiency in insulin**
- Without insulin, the body cannot use glucose as an energy source and must obtain energy via **lipolysis**.
- This process produces **ketones** and leads to **acidosis**.
- DKA occurs frequently in young patients with T1DM on initial presentation

Diabetic Ketoacidosis

Clinical Presentation

- Polyuria, polydipsia, polyphagia
- Weight loss
- Fatigue
- Dyspnea
- Vomiting
- Preceding febrile illness
- Abdominal pain
- Dehydration (tachycardia, poor skin turgor, dry mucous membranes, and orthostatic hypotension)
- Kussmaul's Respiration: Deep, rapid breathing (compensation for **metabolic acidosis**)
- Fruity smell(acetone) on the patient's breath.
- Mental status change (somnolence, lethargy and coma)

Precipitating Factors for DKA

- ✍ Insulin omission – most common cause of DKA
- ✍ New diagnosis of diabetes
- ✍ Infection/sepsis
- ✍ Myocardial infarction
- ✍ Thyrotoxicosis
- ✍ Drugs (glucocorticoids, SGLT2 inhibitors, thiazides, sympathomimetics, cocaine)

Diagnostic Criteria for DKA:

- $\text{pH} \leq 7.3$
- Bicarbonate ≤ 18 mmol/L
- High anion gap > 12 mEq/L (metabolic acidosis)
= (sodium + potassium) – (chloride + bicarbonate)
- Positive serum or urine ketones
- Plasma glucose ≥ 250 mg/dL (but may be lower)
- Precipitating factor

Treatment of DKA

- **Treatment**
 - a) Fluid resuscitation
 - b) Avoidance of hypokalemia
 - c) Insulin administration
 - d) Avoidance of rapidly falling serum osmolality
 - e) Identify and treat precipitating cause

Treatment of DKA

- **Fluid replacement**

✍ The fluid deficit is typically about 100 ml/kg body weight, which amounts to 5 – 7 L in adult patient

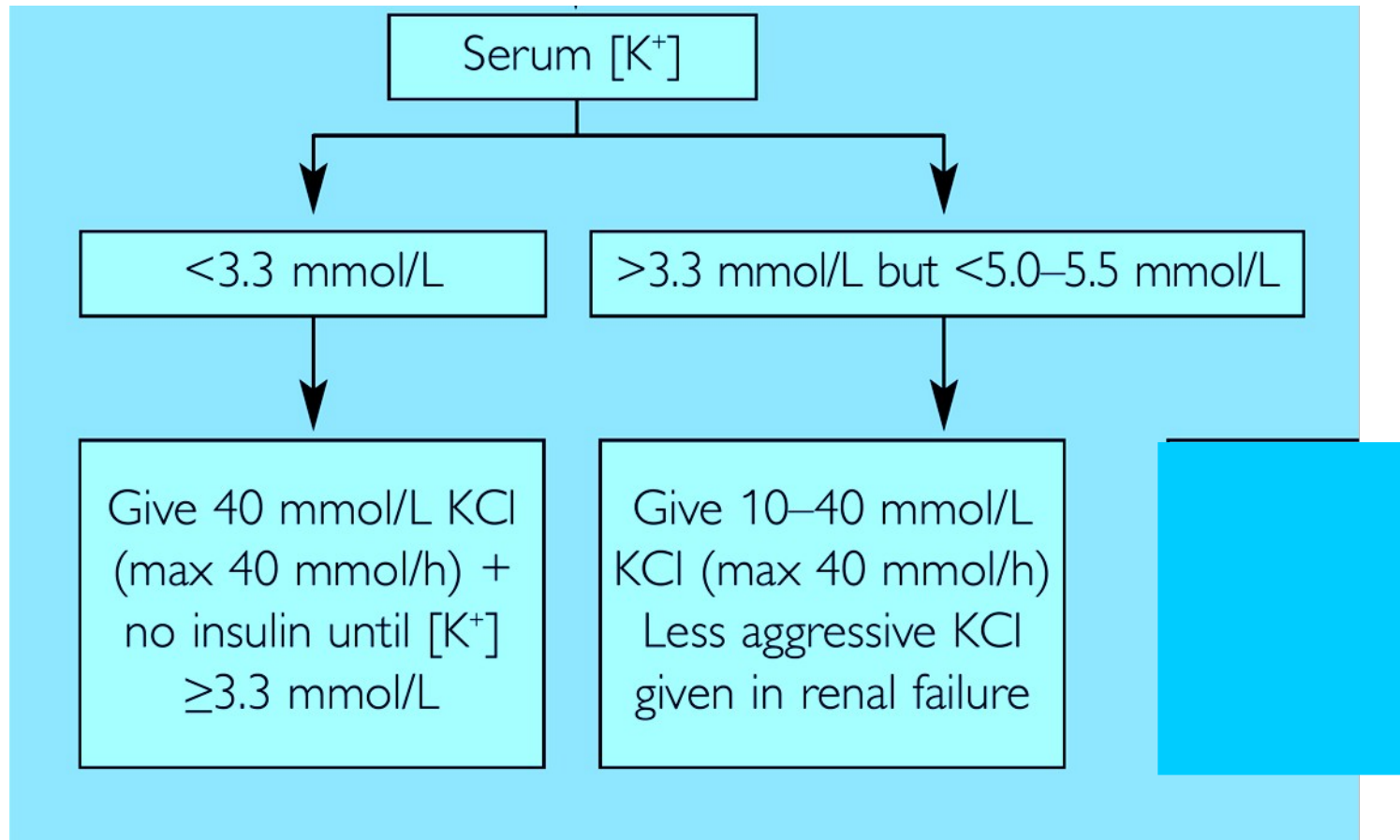
✍ **IV 0.9% sodium chloride** 1 to 1.5 L of fluid within the first hour and thereafter 250 to 500 ml/hour.

✍ Change to **5% dextrose** with NS when serum glucose is less than 200 mg/dL

Goal: Correct dehydration, restore perfusion, and improve insulin response.

Treatment of DKA

- **Potassium:** Correct K^+ first THEN start insulin



Treatment of DKA

- **Insulin**

 First line:

- Intravenous bolus: 0.1 unit/kg
- Intravenous infusion: 0.1 unit/kg/hour
- Continue until acidosis resolves ($\text{pH} > 7.3$, $\text{HCO}_3^- > 18$, Anion gap < 12)

 Alternatively,

- IV infusion of 0.14 unit/kg/hour if no insulin bolus is given

 Reduce infusion rate to 0.02–0.05 unit/kg/hour when serum glucose reaches 200 mg/dL

 Keep glucose between 150 and 200 mg/dL until DKA resolves

Treatment of DKA

- **Bicarbonate Therapy (Only if pH <6.9)**
 - pH 6.9–7.0: 50 mEq NaHCO₃ over 1 hour
 - pH <6.9: 100 mEq NaHCO₃ + KCl in 400 mL sterile water over 2 hours
 - **Monitor pH every 2 hours**
 - Routine bicarbonate therapy is not recommended unless pH <6.9 (severe acidosis)
- **DKA is resolved when:**
 - ✓ **Glucose <200 mg/dL**
 - ✓ **Anion gap <12**
 - ✓ **HCO₃⁻ ≥18 mEq/L**
 - ✓ **pH >7.3**

Hyperosmolar Hyperglycemic State

- HHS occurs primarily in older patients with T2DM
- Dx criteria:
 - plasma glucose value of > 600 mg/dL and
 - serum osmolality of > 320 mOsm/kg
- Treatment:
 - ✍ Aggressive rehydration,
 - ✍ Correction of electrolyte imbalances, and
 - ✍ Continuous insulin infusion to normalize serum glucose.

DKA vs HHS

DKA

- Ketoacidosis
- ECFV contraction
- Milder hyperosmolarity
- Normal to high glucose
- May have ↓ LOC
- Beware hypokalemia
- Must use insulin
- Absolute insulin deficiency + glucagon

HHS

- Minimal acid-base problem
- ECFV contraction
- Hyperosmolarity
- Marked hyperglycemia
- Marked ↓↓ LOC
- Beware hypokalemia
- May need insulin
- Relative insulin deficiency

Treatment of Long-Term Complications

- **Retinopathy**

- ✍ Diabetic retinopathy occurs when the microvasculature that supplies blood to the retina becomes damaged.

- ✍ Leading cause of blindness in adults

- ✍ Routine ophthalmic examination

- ✍ Glucose and blood pressure control

- **Nephropathy**

- ✍ DM is a leading contributor to end-stage renal disease.

- ✍ Urine protein tests annually in T2DM patients.

- ✍ Glycemic control and blood pressure control are primary measures

- ✍ ACE inhibitors and angiotensin II receptor blockers

Neuropathy

- The most common types of peripheral neuropathy are
 1. Chronic sensorimotor distal peripheral neuropathy
 - Pain, tingling, and numbness in the feet and hands,
 2. Autonomic neuropathy
 - Hypotension, gastroparesis, sexual dysfunction, and autonomic failure in response to hypoglycemia
- Treatment:
 - ✍ TCA: amitriptyline, desipramine
 - ✍ Anticonvulsants: gabapentin, lamotrigine, pregabalin
 - ✍ SSRI: paroxetine, citalopram
 - ✍ Opioids: tramadol

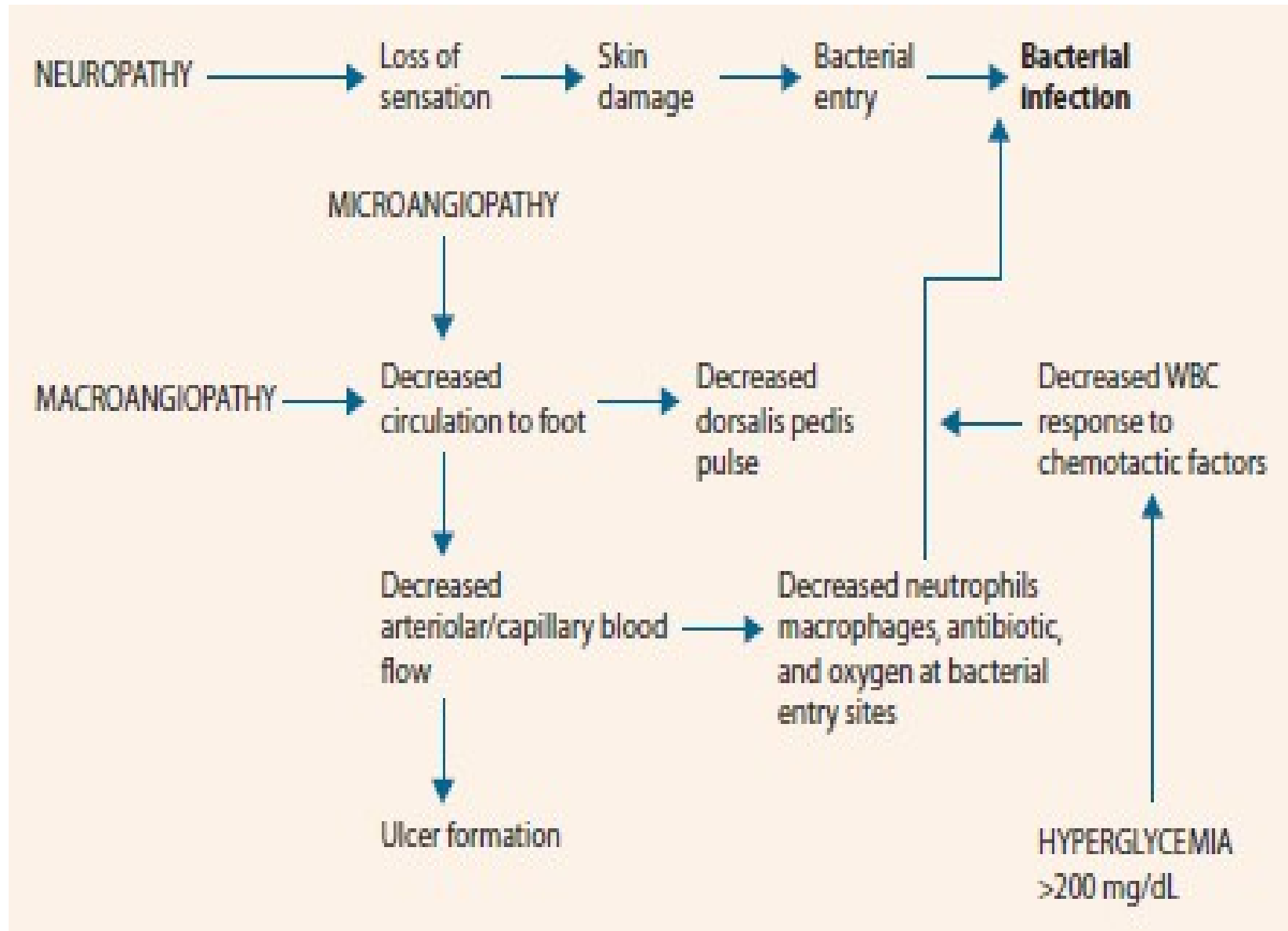
Chronic Complications



- **Diabetic Foot Ulcers**

- **Peripheral arterial disease** causes ischemia to the lower limbs.
- This **decreased blood flow** deprives the tissues of oxygen and nutrients and impairs the ability of the immune system to function adequately
- Loss of sensation in the feet due to neuropathies allows trauma to go unnoticed.
- Daily visual examination of the feet

Pathogenesis



Diabetic Foot Ulcer

- **Management**

- Wound Care & Offloading
- Infection Control
- Glycemic Control
- Revascularization (for PAD)
- Prevention Strategies

Patient Counselling:

1. Avoid activities that can injure the feet
2. Use care when trimming the nails
3. Wash and check the feet daily
4. Choose socks and shoes carefully
5. Routine foot exams

THYROID DISORDERS

Introduction

- Thyroid hormones affect the function of virtually every organ system.
- In a **child**, thyroid hormone is critical for **normal growth and development**.
- In an **adult**, the major role of thyroid hormone is to **maintain metabolic stability**.

Hyperthyroid Disorders (Thyrotoxicosis)

- **Classifications**

- a) Toxic diffuse goiter (Graves' disease):**

- ✍ Most common hyperthyroid disorder
 - ✍ Autoimmune disorder:
 - Thyroid-stimulating **antibodies** directed at thyrotropin receptors mimic thyroid-stimulating hormone (TSH) and stimulate triiodothyronine (T3), thyroxine (T4) production

- b) Pituitary adenomas:**

- ✍ Produce excessive **TSH** secretion that does not respond to normal T3 negative feedback

- c) Toxic adenoma:**

- ✍ Nodule in thyroid, autonomous of pituitary and TSH

e) Toxic multinodular goiter (Plummer disease)

f) Painful subacute thyroiditis:

- Self-limiting inflammation of the thyroid gland caused by **viral invasion** of the parenchyma,
- Result in the release of stored hormone

g) Drug induced

- Excessive exogenous thyroid hormone dosages,
- Amiodarone therapy

Diagnosis

- TSH (suppressed due to excessive T3/T4)
- ↑ Free T4 and/or ↑ T3
- Radioactive iodine uptake scan (to differentiate Graves' disease from thyroiditis)
 - Elevated
 - Graves' disease, TSH-secreting adenoma, toxic adenoma, multinodular goiter
 - Suppressed
 - Caused by thyroiditis or hormone ingestion
- Thyroid antibodies (e.g., TSH receptor antibodies in Graves' disease)

Clinical presentation

- Weight loss or increased appetite
- Stare and Lid lag
- Heat intolerance
- Goiter(thyroid enlargement)
- Fine hair
- Tremor, heart palpitations or tachycardia
- Nervousness, anxiety, insomnia
- Menstrual disturbances (lighter or more infrequent menstruation, amenorrhea) caused by hypermetabolism of estrogen
- Sweating or warm, moist skin
- Exophthalmos, pretibial myxedema in Graves' disease

Clinical presentation



hair loss

HYPERTHYROIDISM

symptoms



bulging eyes



sweating



enlarged thyroid



rapid heartbeat



nervousness



heat intolerance



irritability



tremor of fingers



difficulty sleeping



warm moist palms



scant menstrual period



infertility



muscle weakness



weight loss



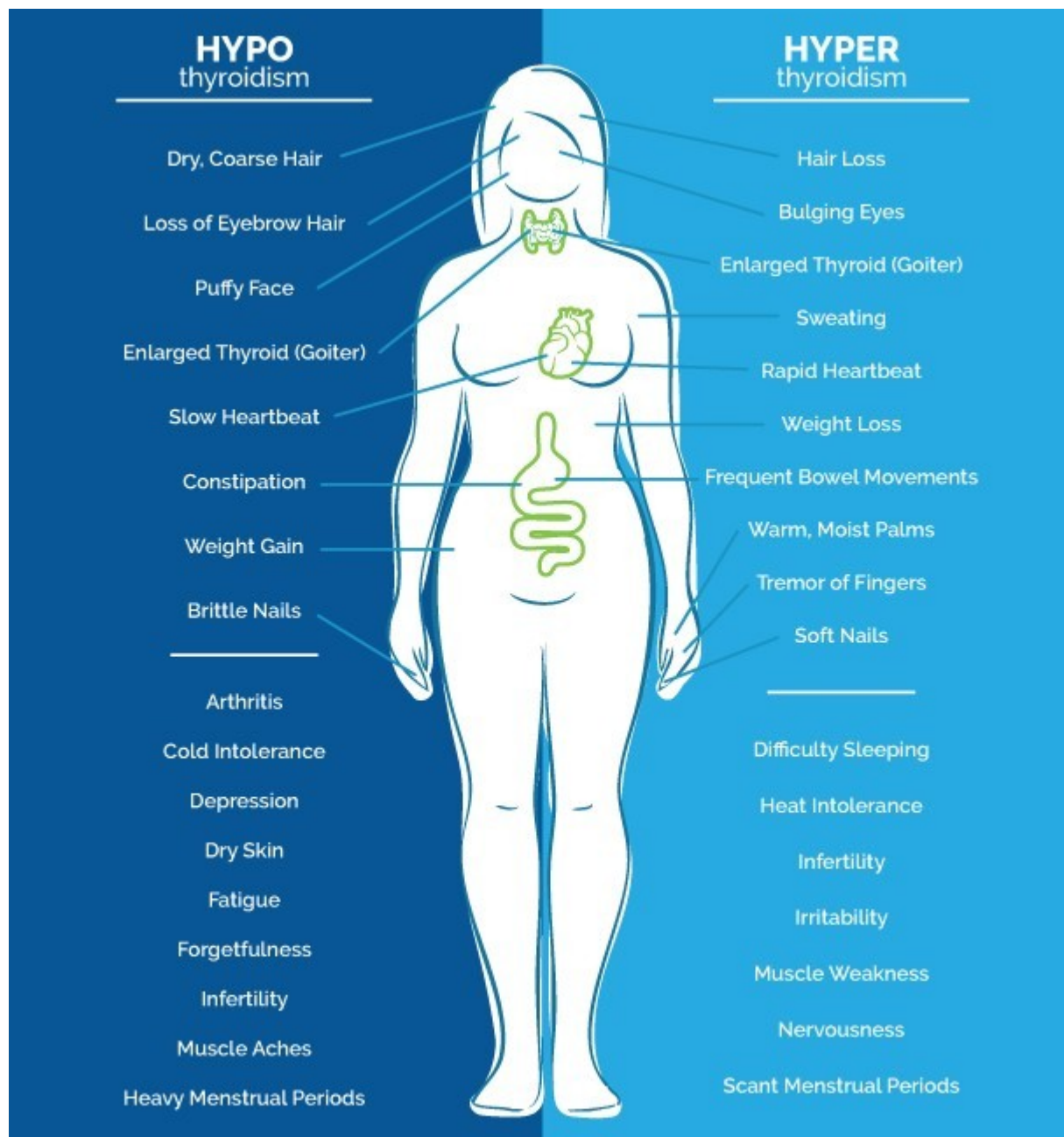
frequent bowel movements



soft nails

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Clinical Presentation



Management of Hyperthyroidism

- **Therapy goals**

- ✍ Minimize or eliminate symptoms, improve quality of life.
- ✍ Minimize long-term damage to organs
 - Heart disease
 - Arrhythmias
 - Sudden cardiac death
 - Bone demineralization
 - Fractures
- ✍ Normalize free T4 and TSH concentrations

Management of Hyperthyroidism

I. Ablative therapy (removal of thyroid tissue):

- ✍ Treatment of choice for Graves' disease, toxic nodule, multinodular goiter
- ✍ According to patient preferences or comorbidities
 - ✍ Radioactive iodine ablation therapy ($I-131$)
 - ✍ Surgical resection (Thyroidectomy) for adenomas
- ✍ Ablative therapy often results in hypothyroidism

Management of Hyperthyroidism

2. Antithyroid pharmacotherapy usually reserved for

- i. Those awaiting ablative therapy or surgical resection
 - (a) Depletes stored hormone
 - (b) Minimizes risk of post-treatment hyperthyroidism caused by thyroiditis
- ii. Those who are not an ablative or surgical candidates
 - a) Serious cardiovascular disease,
 - b) Candidate unlikely to be adherent to radiation safety
- iii. When ablative therapy or surgical resection fails to normalize thyroid function

Management of Hyperthyroidism

- iv. Those with a high probability of remission with oral therapy for Graves' disease:
 - a) Mild disease
 - b) Small goiter
 - c) Low or negative antibody titers
- v. Those with limited life expectancy

Thioureas

- Inhibit iodination and synthesis of thyroid hormones; PTU may block T4/T3 conversion in the periphery as well

(a) Propylthiouracil (PTU)

(1) Preferred agent in pregnant women in the 1st trimester

(2) Dose:

- Initial: 100 mg by mouth three times daily
- Maximal: 400 mg three times daily
- Once euthyroid, may reduce to 50 mg 2-3 times daily

(b) Methimazole

(1) Preferred agent, except in pregnant women (1st trimester)

(2) Dose:

- Initial: 10–20 mg by mouth once daily
- Maximal: 40 mg three times daily
- Once euthyroid, may reduce to 5–10 mg/day

Thioureas

- Adverse effects

-  Hepatotoxicity risk with PTU: Baseline LFT

-  Rash

-  Arthralgias, lupus-like symptoms

-  Fever

-  Agranulocytosis: Baseline CBC

Nonselective β -blockers (propranolol)

- Blocks many hyperthyroidism manifestations mediated by β -adrenergic receptors
 - ✍ Used primarily for **symptomatic relief** (palpitations, tachycardia, tremor, anxiety)
- Also block T4 (less active) conversion to T3 (more active)
- Dosing
 - ✍ Initial: 20–40 mg by mouth three or four times daily
 - ✍ Maximal: 240–480 mg/day
- Alternatives to β -blockers: Clonidine, nondihydropyridine calcium channel blocker

Iodine and iodides

- Inhibits the release of stored thyroid hormone.
- Minimal effect on hormone synthesis.
- Helps decrease vascularity and size of gland before surgery
- Dosing: 120–400 mg split three times daily
 - (a) Lugol's solution (6.3–8 mg iodide per drop)
 - (b) Saturated solution of potassium iodide (38–50 mg iodide per drop)
 - (c) Potassium iodide tablets: (130-mg tablets contain 100 mg of iodide)

Iodine and iodides

- Adverse effects
 - (a) Hypersensitivity
 - (b) Metallic taste
 - (c) Soreness or burning in mouth or tongue
 - (d) Do not use in the days before ablative iodine therapy (may reduce uptake of radioactive iodine).

Thyroid Storm

- Severe and life-threatening decompensated thyrotoxicosis
- Mortality rate may be as high as 20%.
- Precipitating causes:
 - ✍ Trauma
 - ✍ Infection
 - ✍ Antithyroid agent withdrawal
 - ✍ Severe thyroiditis
 - ✍ Postablative therapy (especially if inadequate pretreatment)
- Presentation:
 - ✍ Fever, tachycardia, vomiting, dehydration, coma, tachypnea, delirium

Management of Thyroid Storm

- 1) PTU: 500-1000-mg loading dose, then 250 mg Q4 hours
 Blocks new hormone synthesis
- 2) Alternative: methimazole 60–80 mg daily
- 3) Iodide therapy 1 hour after PTU initiation
- 4) β -Blocker therapy: Propranolol to control symptoms and block conversion of T₄ to T₃
- 5) Paracetamol as antipyretic therapy, if needed (avoid NSAIDs)
- 6) Corticosteroid therapy: Prednisone 300 mg IV loading dose then 100 mg every 8 hours (Dexamethasone, hydrocortisone).
 Prophylaxis against relative adrenal insufficiency and may block conversion of T₄ to T₃.

Case Study

A 43-year-old woman has received a diagnosis of Graves' disease. She is reluctant to try ablative therapy and wants to attempt oral pharmacotherapy first. Her thyroid laboratory values today include TSH 0.22 mIU/L (normal 0.5–4.5 mIU/L) and a free T4 concentration of 3.2 ng/dL (normal 0.8–1.9 ng/dL). She is anxious and always feels warm when others say it is too cold. Which would be considered the best drug for initial treatment of her condition?

- A. Lugols solution
- B. PTU
- C. Atenolol
- D. Methimazole

Hypothyroid Disorders

Classification

- a) **Hashimoto's disease:** Most common hypothyroid disorder in areas with iodine insufficiency
 - ✍ Autoimmune-induced thyroid injury resulting in **decreased thyroid secretion**
 - ✍ **Disproportionately affects women**
- b) **Iatrogenic:** Thyroid resection or radioiodine ablative therapy for treatment of hyperthyroidism
- c) **Iodine deficiency:** most common cause worldwide
- d) Congenital Hypothyroidism (Cretinism)

Secondary causes

- (i) **Pituitary insufficiency** (failure to produce adequate TSH secretion, **central** or **secondary hypothyroidism**)
- (ii) **Drug induced** (amiodarone, lithium)

Clinical Presentation

- ✍ Cold intolerance
- ✍ Dry skin
- ✍ Fatigue, lethargy, weakness
- ✍ Weight gain
- ✍ Bradycardia
- ✍ Slow reflexes
- ✍ Coarse skin and hair
- ✍ Periorbital swelling
- ✍ Menstrual disturbances (more frequent or longer menstruation, painful menstruation, menorrhagia) caused by hypometabolism of estrogen
- ✍ Goiter (primary hypothyroidism)

Diagnosis of Hypothyroidism

- Primary Hypothyroidism: ↑TSH, ↓ Free T4
- Secondary Hypothyroidism: ↓TSH, ↓ Free T4
- Thyroid antibodies (elevated in Hashimoto's thyroiditis)

Management of Hypothyroidism

- Therapy goals

- ✍ Minimize or eliminate symptoms; improve quality of life.
- ✍ Minimize long-term damage to organs (myxedema coma, heart disease).
- ✍ Normalize free T4 and TSH concentrations

Management of Hypothyroidism

Therapeutics

(a) **Levothyroxine** (drug of choice)

 Mechanism of action: Synthetic T4

 Dosing

✓ Initial

 In healthy adults, 1.6 mcg/kg per day

 In patients 50–60 years of age, consider 50 mcg/day.

 In those with existing cardiovascular disease, consider 12.5–25 mcg/day

✓ Dosage titration based on response

 Can increase or decrease in 12.5- to 25-mcg/day increments

Management of Hypothyroidism

- **Adverse effects**

- (a) Hyperthyroidism
- (b) Cardiac abnormalities (tachyarrhythmias, angina, myocardial infarction)
- (c) Linked to risk of fractures (usually at higher dosages or over supplementation)

- **Efficacy**

-  Considered drug of choice because of its adverse effect profile, cost, lack of antigenicity, and uniform potency

Myxedema Coma

- Occurs in long-standing untreated or inadequately treated hypothyroidism
- Severe and life-threatening decompensated hypothyroidism.
- Mortality rate 30%–60%
- Precipitating causes: Trauma, infections, heart failure, medications (e.g., sedatives, narcotics, anesthesia, lithium, amiodarone)
- Presentation: Coma is uncommon, despite terminology; altered **mental state** (very common); diastolic hypertension; **hypothermia**; hypoventilation, **bradycardia**

Myxedema Coma

Pharmacotherapy

(a) Intravenous thyroid hormone replacement

- T4: 100- to 500-mcg loading dose, followed by 75–100 mcg/day,
- T3 may be preferred over T4,
 - T3 is more biologically active
 - T4/T3 conversion may be suppressed in myxedema coma
 - Cost and availability limit intravenous T3

(b) Antibiotic therapy: empiric therapy with broad spectrum antibiotics

(c) Corticosteroid therapy

- ↳ Hydrocortisone 100 mg every 8 hours (or equivalent steroid)

Case Study

A 63-year-old woman has Hashimoto's disease. Her thyroid laboratory values today include the following: TSH 10.6 mIU/L (normal 0.5–4.5 mIU/L) and a free T4 concentration of 0.5 ng/dL (normal 0.8–1.9 ng/dL). She feels consistently run down and has dry skin that does not respond to the use of hand creams. Which would be considered the best drug for initial treatment of her condition?

- A. Levothyroxine.
- B. Liothyronine.
- C. Desiccated thyroid.
- D. Methimazole.

Thank You